

Data Flow Diagram

Date: September 2026

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Project Name: EduGenie – Google Gemini Powered Learning Assistant

Maximum Marks: 2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) shows how data moves through EduGenie between the User, the main processes and the Google Gemini-powered AI service. The system receives learning requests and returns AI-generated educational content.

Processes and Data Sources

No.	Process	What it does
1	User Request – Select Learning Feature	Receives the user's selected feature and prompt.
2	Input Validation – Validate Request	Checks the entered prompt and required input.
3	AI Processing – Generate Learning Response	Sends the validated request to the Google Gemini-powered service.
4	Explanation / Q&A – Understand Concept	Returns explanations and answers for learning questions.
5	Quiz & Summary – Generate Practice Content	Generates quizzes or summarizes user-provided text.
6	Learning Path – Recommend Learning Steps	Creates learning recommendations based on topic and level.

Data Sources

Data Source	Contents
User Input	Prompt, question, text, topic and learning level.
Google Gemini API / AI Service	AI-generated explanations, answers, quizzes, summaries and learning recommendations.
Learning Request Data	Validated request details passed to the appropriate process.
Generated Response	Final educational content displayed to the user.

EduGenie – Data Flow Diagram (Level 1)

Data Flow	Movement
User → Process 1	Feature selection, prompt / text
Process 1 → Process 2	Learning request
Process 2 → Process 3	Validated request
Process 3 ↔ Google Gemini	Prompt / generated AI content
Process 3 → Processes 4–6	Required learning request
Processes 4–6 → User	Explanation, answer, quiz, summary or learning path

Note: EduGenie uses the Google Gemini-powered AI service to generate learning content. The User is the main external entity.